SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Capstone project presentation

Smart healthcare resource allocation and appointment scheduling system

Team - 07

CSA4305 –Internet Programming

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INTRODUCTION

- A Smart Hospital Resource and Appointment Management Portal is a web-based system for managing hospital activities.
- It helps patients book appointments with doctors quickly and easily.
- The system stores patient, doctor, and hospital resource information in one place.
- It reduces paperwork, saves time, and avoids scheduling conflicts.
- It improves hospital efficiency and provides better service to patients.

OBJECTIVES

- Develop a centralized system to manage patients, doctors, appointments, and hospital resources.
- Automate appointment scheduling to reduce waiting time and booking conflicts.
- Improve the management of beds, rooms, equipment, and staff availability.
- Provide secure records, real-time monitoring, and reports for better hospital management.
- Enhance the overall patient experience by providing faster, reliable, and efficient healthcare services.

PROBLEM STATEMENT

- Hospitals often use manual or separate systems to manage patient records and appointments.
- Appointment scheduling conflicts can lead to long waiting times and inconvenience for patients.
- Poor management of beds, rooms, and medical equipment reduces hospital efficiency.
- Lack of centralized data makes it difficult to track patients, doctors, staff, and resources.
- A centralized hospital management portal is needed to improve efficiency, accuracy, and the quality of healthcare services.

Literature Review

No	Year	Article Name	Reference	Focus	Methods	Result
01	2025	*Patient Flow Optimization in Hospitals Using Queueing Theory*	T. Wang	Improving patient flow to reduce overcrowding and delays	Queueing models, simulation of OPD and emergency services	Reduced patient waiting time and better utilization of hospital resources
02	2025	*Impact of Online Appointment Systems on Patient Satisfaction*	P.K. Betancor et al.	Evaluating effectiveness of online hospital appointment booking	Surveys and staff–patient feedback analysis	Higher patient satisfaction and fewer missed appointments
03	2025	*Privacy-Preserving Appointment Scheduling in Smart Healthcare Systems*	S. Zhang	Ensuring secure and privacy-safe hospital appointment booking	Encryption techniques, secure login	Improved data security and increased patient trust
04	2024	*Design and Development of a Smart Hospital Management System Portal*	G. Gaikwad	Developing a web-based smart HMS portal	PHP, MySQL database, role-based web system implementation	Faster processing, efficient data handling, and reduced paperwork
05	2025	*Global Market Trends in Patient Flow and Hospital Digital Solutions*	RootsAnalysis	Analyzing adoption of digital patient-flow and HMS technologies	Market statistics, industry reports, and trend analysis	Growth in digital healthcare adoption and smart hospital investments

Module 1
Patient Appointment Management

Aim:

To provide an efficient system for managing patient appointments, records, and doctor scheduling in a centralized platform.

Key Points:

- ➢ Register and manage patient records.
- ➢ Book appointments with available doctors.
- ➢ Reschedule or cancel appointments.
- ➢ View doctor consultation schedules.
- ➢ Send appointment reminder notifications.

Output:

A streamlined appointment management system that reduces waiting time and improves patient service.

Module 2
Doctor & Staff Management

Aim:

To efficiently manage doctor and staff information, schedules, attendance, and availability for smooth hospital operations.

Key Points:

- Maintain doctor and staff profiles.
- Create duty schedules and shifts.
- Track attendance and leave records.
- Monitor doctor availability regularly.
- Prevent scheduling conflicts efficiently.

Output:

An organized workforce management system that improves staff coordination and operational efficiency.

Module 3
Hospital Resource Management

Aim:

To monitor and manage hospital resources effectively for better utilization and uninterrupted healthcare services.

Key Points:

- Monitor beds and room availability.
- Allocate resources based on needs.
- Track medical equipment usage.
- Schedule equipment maintenance regularly.
- Generate resource utilization reports.

Output:

- An efficient resource management system that optimizes hospital facilities and improves service quality.

Module 4
Administration & Analytics Dashboard

Aim:

To provide administrators with centralized monitoring, reporting, and analytical tools for effective hospital management.

Key Points:

- ➢ Monitor overall hospital activities.
- ➢ Manage users and permissions securely.
- ➢ Generate reports and analytics.
- ➢ Display real-time performance dashboards.
- ➢ Support better administrative decisions.

Output:

A centralized dashboard that provides real-time insights and supports better decision-making in hospital administration.

CONCLUSION

- The portal provides a centralized platform for managing hospital operations efficiently.
- It simplifies patient appointments, doctor schedules, and hospital resource management.
- The system reduces paperwork, improves accuracy, and saves valuable time.
- Overall, it enhances hospital efficiency and delivers better healthcare services to patients.

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Engineer to Excel

Thank You